

Geometric Brownian Motion: Exact Terminal Sampler

hasklib mathematical companion

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Abstract

This note derives the result that `GBM::sample_terminal` relies on: applying Itô's lemma to $\ln X_t$ linearises the geometric Brownian motion SDE and yields an exact (not discretised) sampler for X_T given X_0 . We also derive the first two moments, which are what `test_stochastic.cpp` checks against simulated output. This file assumes Itô's lemma as established in `ito_foundations.tex` (Theorem 1 there); it is not re-derived here. The derivation is classical; see ? for the standard textbook treatment.

1 The GBM SDE

Definition 1 (Geometric Brownian motion). X_t follows geometric Brownian motion with drift $\mu \in \mathbb{R}$ and volatility $\sigma > 0$ if it solves

$$dX_t = \mu X_t dt + \sigma X_t dW_t, \quad X_0 > 0. \quad (1)$$

In the notation of `ito_foundations.tex` this is the choice $a(t, x) = \mu x$, $b(t, x) = \sigma x$ in the general SDE $dX_t = a(t, X_t)dt + b(t, X_t)dW_t$.

Remark 1 (Correspondence with the code). This is exactly the pair of functions `GBM::drift` and `GBM::diffusion` return: `drift(t,x) =  $\mu x$` , `diffusion(t,x) =  $\sigma x$` . `EulerMaruyama` consumes these polymorphically through the `StochasticProcess` interface; the sampler derived below is the *exact* alternative to that discretisation, used by `GBM::sample_terminal` instead of stepping the Euler scheme forward.

Both a and b are linear in x and (for $x > 0$, which GBM never leaves since $X_t = X_0 \exp(\cdot) > 0$ once we derive the solution below) globally Lipschitz on any bounded interval away from 0, so the existence–uniqueness assumption of `ito_foundations.tex` applies and (1) has a unique strong solution.

2 Linearising via Itô's lemma

The SDE (1) has no constant-coefficient closed form as written, because both coefficients scale with X_t . The standard fix is to apply Itô's lemma to $f(t, x) = \ln x$: this removes the x -dependence from both coefficients and turns (1) into an SDE with *constant* drift and diffusion, which integrates directly.

Theorem 1 (Exact terminal law of GBM). *Let X_t solve (1). Then for any $T > 0$,*

$$X_T = X_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T\right), \quad (2)$$

where $W_T \sim \mathcal{N}(0, T)$.

Proof. Apply Itô's lemma (Theorem 1 of `ito_foundations.tex`) to $f(t, x) = \ln x$, so $f_t = 0$, $f_x = 1/x$, $f_{xx} = -1/x^2$. With $a(t, X_t) = \mu X_t$ and $b(t, X_t) = \sigma X_t$,

$$d(\ln X_t) = \left(0 + \mu X_t \cdot \frac{1}{X_t} + \frac{1}{2} \sigma^2 X_t^2 \cdot \left(-\frac{1}{X_t^2}\right)\right) dt + \sigma X_t \cdot \frac{1}{X_t} dW_t.$$

The X_t factors cancel completely — this cancellation is the whole point of the $\ln$ substitution — leaving

$$d(\ln X_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t. \quad (3)$$

The right-hand side no longer depends on X_t : it is Brownian motion with constant drift $\mu - \frac{1}{2}\sigma^2$ and constant volatility σ , so (3) integrates directly from 0 to T :

$$\ln X_T - \ln X_0 = \left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma(W_T - W_0) = \left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T,$$

using $W_0 = 0$. Exponentiating gives (2). Since $W_T \sim \mathcal{N}(0, T)$ by definition of Brownian motion, this is a complete description of the law of X_T given X_0 . $\square$

Remark 2. Equation (2) is sampled with a single normal draw: generate $Z \sim \mathcal{N}(0, 1)$, set $W_T = \sqrt{T} Z$, and evaluate the right-hand side. No time-stepping is needed — this is what makes it an *exact* sampler rather than a discretisation, and it is exactly what `GBM::sample_terminal` implements using `NormalRng::draw()`.

Remark 3 (Where the $-\frac{1}{2}\sigma^2$ comes from). The drift that appears in (2) is *not* μ , the drift of X_t itself, but $\mu - \frac{1}{2}\sigma^2$. That correction is exactly the $\frac{1}{2}b^2 f_{xx}$ term of Itô's lemma — the term with no analogue in ordinary calculus, discussed at length in `ito_foundations.tex`. If one (incorrectly) applied the ordinary chain rule to $\ln X_t$, the correction would be missing and the resulting sampler would be biased upward in expectation, as proposition 1 below makes precise.

3 Moments

The moment-matching test in `test_stochastic.cpp` checks simulated terminal samples against the following closed-form mean and variance.

Lemma 1 (Moment generating function of a Gaussian). *If $Y \sim \mathcal{N}(m, v)$ then $\mathbb{E}[e^Y] = \exp(m + \frac{1}{2}v)$.*

Proof. Write $Y = m + \sqrt{v} Z$ with $Z \sim \mathcal{N}(0, 1)$. Complete the square in the Gaussian integral:

$$\mathbb{E}[e^Y] = e^m \mathbb{E}[e^{\sqrt{v} Z}] = e^m \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sqrt{v} z - \frac{1}{2} z^2} dz = e^m e^{\frac{1}{2}v} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sqrt{v})^2} dz = e^{m + \frac{1}{2}v},$$

the last integral being 1 since it is a $\mathcal{N}(\sqrt{v}, 1)$ density integrated over $\mathbb{R}$. $\square$

Proposition 1 (Mean of GBM). $\mathbb{E}[X_T] = X_0 e^{\mu T}$.

Proof. By theorem 1, $X_T = X_0 e^Y$ with $Y = (\mu - \frac{1}{2}\sigma^2)T + \sigma W_T$. Since $W_T \sim \mathcal{N}(0, T)$, $Y \sim \mathcal{N}((\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$. Applying lemma 1 with $m = (\mu - \frac{1}{2}\sigma^2)T$ and $v = \sigma^2 T$,

$$\mathbb{E}[X_T] = X_0 \mathbb{E}[e^Y] = X_0 \exp\left((\mu - \frac{1}{2}\sigma^2)T + \frac{1}{2}\sigma^2 T\right) = X_0 e^{\mu T}.$$

The $-\frac{1}{2}\sigma^2 T$ Itô correction and the $+\frac{1}{2}\sigma^2 T$ from the moment generating function cancel exactly, leaving the mean growing at the "natural" rate μ despite the correction appearing inside every individual path. $\square$

Proposition 2 (Variance of GBM). $\text{Var}(X_T) = X_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1)$.

Proof. $\text{Var}(X_T) = \mathbb{E}[X_T^2] - (\mathbb{E}[X_T])^2$. For the second moment, write $X_T^2 = X_0^2 e^{2Y}$ with Y as above, so $2Y \sim \mathcal{N}(2(\mu - \frac{1}{2}\sigma^2)T, 4\sigma^2 T)$. By lemma 1 with $m = 2(\mu - \frac{1}{2}\sigma^2)T$ and $v = 4\sigma^2 T$,

$$\mathbb{E}[X_T^2] = X_0^2 \exp\left(2\left(\mu - \frac{1}{2}\sigma^2\right)T + 2\sigma^2 T\right) = X_0^2 e^{(2\mu + \sigma^2)T}.$$

Subtracting $(\mathbb{E}[X_T])^2 = X_0^2 e^{2\mu T}$ from proposition 1 and factoring,

$$\text{Var}(X_T) = X_0^2 e^{(2\mu + \sigma^2)T} - X_0^2 e^{2\mu T} = X_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1). \quad \square$$

Remark 4 (What the test checks). `test_stochastic.cpp` draws a large number of terminal samples from `GBM::sample_terminal`, computes their sample mean and sample variance, and checks these against propositions 1 and 2 to within Monte Carlo tolerance. This is a check on the *sampler*, not on (1) itself — it would catch, for instance, a sign error in the $-\frac{1}{2}\sigma^2$ correction, since that error changes the mean but leaves the qualitative shape of the distribution intact.